

JavaScript

What is Web Technology?

- ☞ Web technology is the mechanism which enables two or more computing devices to communicate over the Internet in a network.
- ☞ These technology is the various tools and techniques that are utilized in the process of communication between different types of devices over the Internet.
- ☞ Web technology includes the following factors:
 - i. The basics, which will cover web browsers and some web app development fundamentals.
 - ii. Programming language and frameworks which are used in the development of websites.
 - iii. Database that are used at the backend to store data required or collected by website.
 - iv. Some protocols, that is rules for communicating on the web.
 - v. It uses graphics, audiovisual, and other multimedia elements.
 - vi. Some data formats that are usually used to transmit data over the Internet.

Server side and Client Side Scripting

Server side Scripting

- ☞ Server side scripting is a method of designing websites so that the process or user request is run on the originating server.
- ☞ It provides an interface to the user and limit access to proprietary data and help keep control of the script source code.
- ☞ Some language used to create server side scripting are as follows: ASP, SSJS(Server-side JavaScript), PHP, Perl, Ruby, Python, Java etc.

Client Side Scripting

- ☞ Client side scripting is performed to generate a code that can run on the client end i.e. browser without needing the server-side processing. Basically, these types of scripts are placed inside an HTML document.
- ☞ The client side scripting can be used to examine the user's form for the errors before submitting it and for changing the content according to the user input. The Web requires three elements for its functioning which are: client, database and server.
- ☞ The effective client side scripting can significantly reduce the server load. It is design to run as a scripting language utilizing a web browser as a host program. For example, when a user makes a request via browser for a

webpage to the server, it just sent the HTML and CSS as plain text and the browser interprets and renders the web content in the client end.

Introduction to Internet Technology

- ☞ Computer Internet technology refers to devise, software, hardware and transmission protocols used to connect computers together in order to receive or send data from one computer to another within a small network or as part of a small network within a larger network such as the Internet. Included are Local Area Networks (LANs), Wide Area Networks (WANs), bridging, switching, routing, voice and data integration, wireless integration, dial-up technology, cable access technology, network security and network management.

Introduction to JavaScript

- ☞ JavaScript is a lightweight object-oriented programming language, which is used by several websites for scripting the webpages.
- ☞ It is an interpreted, full-fledged programming language that enables dynamic interactivity on websites when applied to an HTML document.
- ☞ Using JavaScript, users can build modern web applications to interact directly without reloading the page every time.

Features of JavaScript

- ☞ Some important features of JavaScript are as follows:
 - I. All popular web browsers support JavaScript as they provide built-in execution environments.
 - II. It is a structured programming language. It follows the syntax and structure of the C programming language.
 - III. JavaScript is an object oriented programming language that uses prototypes rather than using classes for inheritance.
 - IV. It is light-weighted and interpreted language.
 - V. It is a case-sensitive language.
 - VI. JavaScript is supportable in several operating systems including, Windows, macOS etc.
 - VII. It provides good control to the users over the web browsers.

Use of JavaScript

- ☞ JavaScript is used to create interactive websites. It is mainly used for:
 - I. Client-side validation.
 - II. Dynamic drop-down menus.
 - III. Displaying date and time.
 - IV. Displaying pop-up windows and dialog boxes.

V. Displaying clocks.

How to add JavaScript to HTML

- ☞ JavaScript in a page will be executed immediately while the page loads into the browser. This is not always what we want. Sometimes we want to execute a script when a page loads, other times when a user triggers an event.
 - ❖ JavaScript in body section will be executed **WHILE** the page load.
 - ❖ JavaScript in the head section will be executed when **CALLED**.

Script in the body Section

- ☞ Script to be executed when the page loads go in the body section. When you place a script in the body section it generate the content of the page.

Syntax:

```
<HTML>
<HEAD><TITLE></TITLE></HEAD>
<BODY>
<script type="text/javascript">
```

```
</script>
```

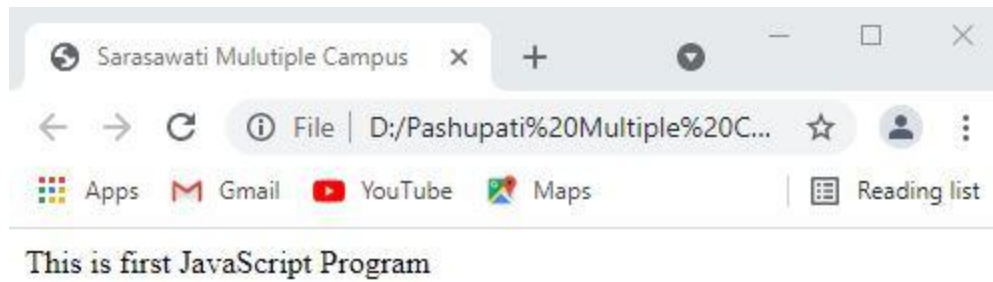
```
</BODY>
</HTML>
```

Example:

```
<HTML>
<HEAD><TITLE></TITLE></HEAD>
<BODY>
<script type="text/javascript">
document.write("This is first JavaScript Program");
</script>

</BODY>
</HTML>
```

Out Put:



JavaScript in a Head Section

- ☞ Script to be executed when they are called, or when an event is **triggered**, go in the head section. When you place script in the head section, you will ensure that the script is loaded before anyone uses it.
- ☞ Script that contains functions go in the head section of the document. Then we can be sure that the script is loaded before the function is called.

Syntax:

```
<HTML>
<HEAD>

<TITLE></TITLE>
<script type="text/javascript">

</script>

</HEAD>
<BODY>

</BODY>
</HTML>
```

Example:

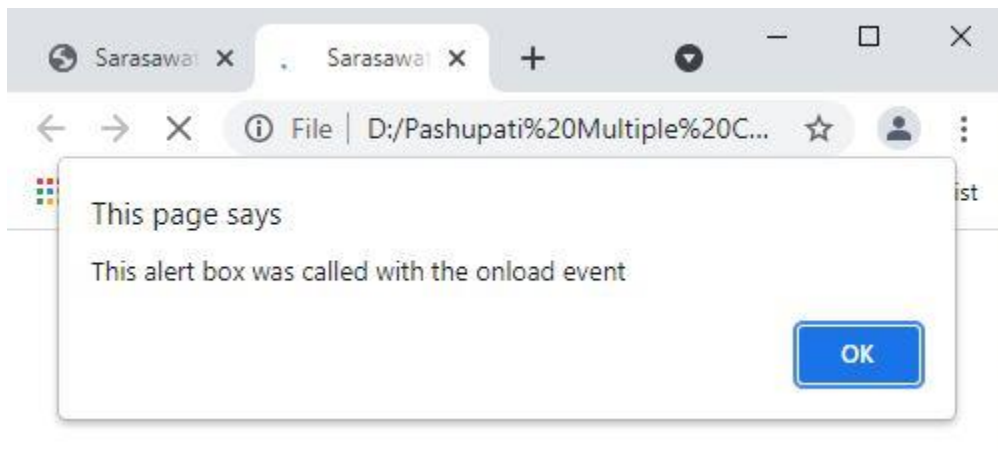
```
<HTML>
<HEAD>
<TITLE></TITLE>
<script type="text/javascript">
function message()
{
```

```
        alert("This is Alert box was called with the onload event");
    }
</script>

</HEAD>
<BODY onload="message()">

</BODY>
</HTML>
```

Out Put:



Note: If you press **OK** then page load will be completed.

Script in the both body and head section

- ☞ You can place an unlimited number of script in your document, so you can have script in both the body section and head section.

Syntax:

```
<HTML>
<HEAD>
<TITLE></TITLE>
<script type="text/javascript">

</script>

</HEAD>
<BODY>
<script type="text/javascript">
```

```
</script>
```

```
</BODY>
```

```
</HTML>
```

How to use external JavaScript

- ☞ Sometime you might want to run the same JavaScript on several pages, without having to write the same script on every page.
- ☞ To simplify this, you can write a JavaScript in an external file. Save the external JavaScript file with a **.js** file extension. Eg. Script.js

Syntax:

```
<HTML>  
<HEAD>  
<TITLE></TITLE>  
<script src="xxx.js">  
</script>
```

```
</HEAD>  
<BODY>  
</BODY>  
</HTML>
```

Note:

- ❖ JavaScript is a case sensitive
- ❖ JavaScript takes whitespace character.

Comments in JavaScript

- ❖ Single line comment
- ☞ You can add comment to your script by using two slashes //(forward slashes).

Example:

```
<HTML>  
<HEAD><TITLE></TITLE></HEAD>  
<BODY>  
<script type="text/javascript">  
//document.write("This is first JavaScript Program\This message is written when page  
load");  
</script>
```

```
</BODY>
</HTML>
```

Out Put:

```
<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Transitional//EN" "http://www.w3.org/TR/xhtml1/DTD/xhtml1-transitional.dtd">
<html xmlns="http://www.w3.org/1999/xhtml"><head>
<meta http-equiv="Content-Type" content="text/html; charset=utf-8" />
<title>Sarasawati Mulutiple Campus</title>

</head>

<body>
<script type="text/javascript">
//document.writeln("This is first JavaScript program\ This message is written when the page is loaded");
</script>
</body>
</html>
```

❖ Multiline Comment

☞ Multiline comment start with /* and end with */

```
</head>

<body>
<script type="text/javascript">
/*
document.writeln("This is first JavaScript program\ This message is written when the page is loaded");
document.writeln("This is first JavaScript program\ This message is written when the page is loaded");
document.writeln("This is first JavaScript program\ This message is written when the page is loaded");
document.writeln("This is first JavaScript program\ This message is written when the page is loaded");
*/
</script>
</body>
</html>
```

JavaScript Variables

☞ Variable is a container for information you want to store. A variable value can change during the script. You can refer to a variable by name to see its value or to change its value.

Rules for Variable name

- ❖ Variable names are case Sensitive. Eg Name, name, NAME all are different
- ❖ They must begin with a letter or the underscore character. Eg Total, total_marks etc.

Declaration of Variable

☞ Variable can be declare with **var** statement.

Syntax:

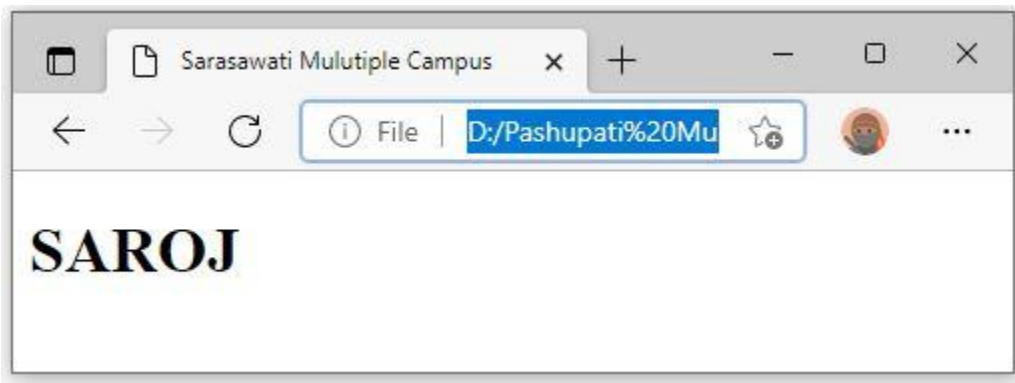
var variable_name = some value;
eg. var name="SAROJ"

Example:

```
<HTML>
<HEAD>
<TITLE></TITLE>

</HEAD>
<BODY>
<script type="text/javascript">
var name="SAROJ";
document.write("<h1>" + name + "</h1>");
</script>

</BODY>
</HTML>
Out Put:
```

**NOTE:**

You can also create a variable without the **var** statement.

Eg. name="SAROJ"

Example:

Example:

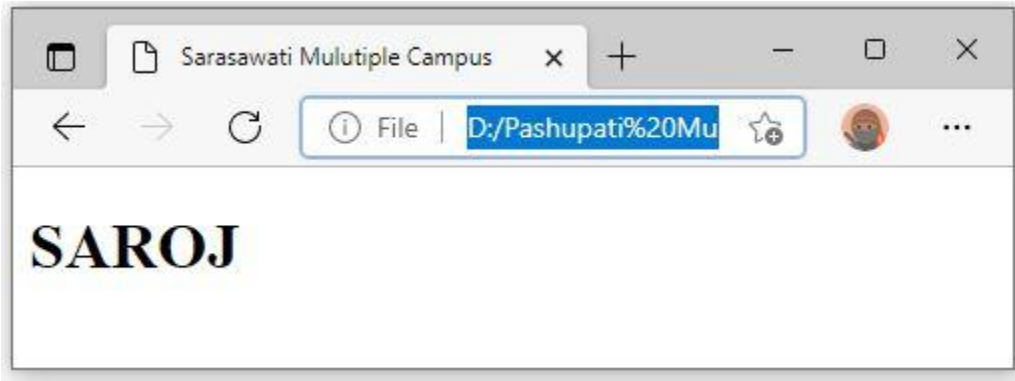
```
<HTML>
<HEAD>
<TITLE></TITLE>

</HEAD>
<BODY>
```

```
<script type="text/javascript">
name="SAROJ";
document.write("<h1>" + name + "</h1>");
</script>

</BODY>
</HTML>
```

Out Put:



Types of JavaScript Variable

1. Local Variable

- ✎ A variable which is declared inside the block or function is called local variable.
- ✎ It is accessible within the function or block only.

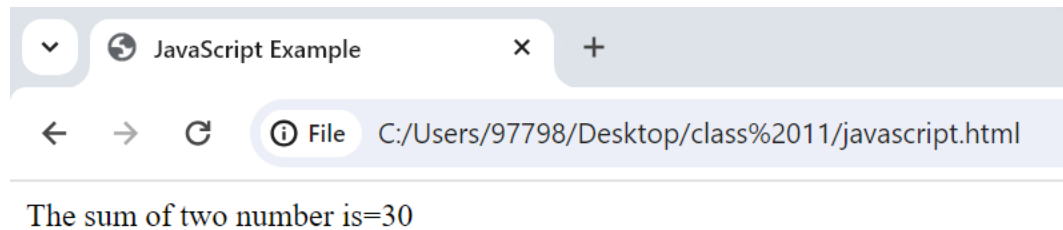
Example:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>JavaScript Example</title>
</head>
<body>
  <script>
    function sum(){
      var x=10;
      var y=20;
      var z=x+y;
      document.write("The sum of two number is="+z);
    }
    sum();
  </script>
```

```
</body>
```

```
</html>
```

Output:



2. Global Variable

- ✎ A JavaScript global variable is accessible from any function.
- ✎ A variable i.e. declared outside the function or declared with a window object is known as a global variable.

Example:

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
<head>
```

```
  <meta charset="UTF-8">
```

```
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
```

```
  <title>JavaScript Example</title>
```

```
</head>
```

```
<body>
```

```
  <script>
```

```
    var x=30;
```

```
    var y=20;
```

```
    function sum(){
```

```
      var z=x+y;
```

```
      document.write("The sum of two number is="+z);
```

```
      document.write("<br>");
```

```
    }
```

```
    function sub(){
```

```
      var z=x-y;
```

```
      document.write("The difference of two number is="+z);
```

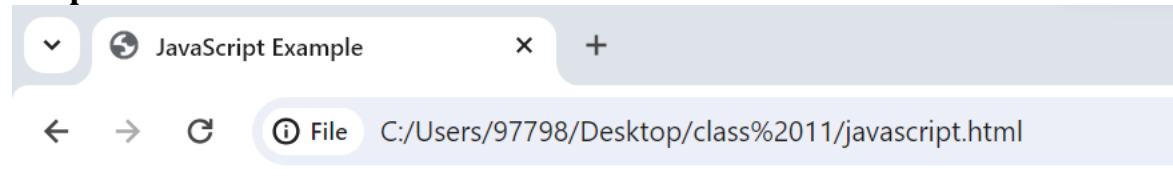
```
    }
```

```
    sum();
```

```
    sub();
```

```
</script>
</body>
</html>
```

Output:



The sum of two number is=50
The difference of two number is=10

JavaScript Data Type

- ✎ Data types basically specify what kind of data can be stored and manipulated within program.
- ✎ JavaScript provides different data types to hold different types of values.
- ✎ There are two types of data types in JavaScript.
 1. Primitive data type.
 2. Non-primitive (reference) data type.

1. Primitive Data Type

- ✎ Primitive data types can hold only one value at a time.
- ✎ There are five types of primitive data types in JavaScript.
- ✎ They are as follow:

Data Types	Description
String	It represents a sequence of Characters eg "Hello"
Number	It represents numeric values e.g. 4, 20
Boolean	It represents Boolean value either false or true.
Undefined	It represents an undefined value.
Null	It represents null i.e. no value at all.

2. Non-primitive data type

- ✎ Non-primitive data types can hold collections of values and more complex entities. The non-primitive data types are as follows:

Data Types	Description
Object	It represents an instance through which we can access members.
Array	It represents group of similar values.
RegExp	It represents regular expression.

JAVA SCRIPT OPERATORS

- ☞ Operator is a symbol which performs operation between two or more operand.
- ☞ The different types of operator used in JavaScript are:
 1. Arithmetic Operator
 2. Assignment Operator
 3. Comparison Operator
 4. Logical Operators
 5. String Operator
 6. Conditional Operator

1. Arithmetic Operator

Operator	Description	Example	Result
+	It is used to perform addition operation	X=5 Y=3 Z=x+y	8
-	It is used to perform subtraction operation	Z=x-y	2
*	It is used to perform multiplication operation	Z=x*y	15
/	It is used to perform division operation	Z=x/y	1
%	It is used to perform mode or remainder	Z=x%y	2
++	It increment the value by 1	X=2 X++	3
--	It increment the value by 1	X=3 x--	2

2. Assignment Operator

Operator	Example	Is the same as
=	X=y	X=y
+=	X+=y	X=x+y
-=	x-=y	X=x-y
=	X=y	X=x*y
/=	x/=y	X=x/y
%=	X%=y	X=x%y

3. Comparison Operator

Operators	Description	Example
==	It is used to check equality	(x==y),x=5,y=6, returns false
===	Is equal to(checks for both value and type)	X=5 Y="5" X==y , returns true X===y, returns false
!=	Is not equal to	5!=6, returns true
>	Is greater than	5>4 , it returns true
>=	Is greater than equal to	5>=6, it returns false
<	Is less than	5<6, it returns true
<=	Is less than or equal to	5<=6, it returns true

4. Logical Operators

Operators	Description	Example
&&	Logical AND	X=6,y=3 (x<10 && y>1) returns true
	Logical OR	X=6, y=3 (x==6 y==5) returns true
!	NOT	X=6,y=3 !(x==y) returns true

String Operators

☞ String operator is most often text, for example "Hello World". To stick two or more string variables together, use the + operator.

Example:

Txt1="What a very"

Txt2="nice day"

Txt3=Txt1+Txt2

OR

Txt3=Txt1 +" "+Txt2

Example:

<body>

<script type="text/javascript">

txt1="What a very"

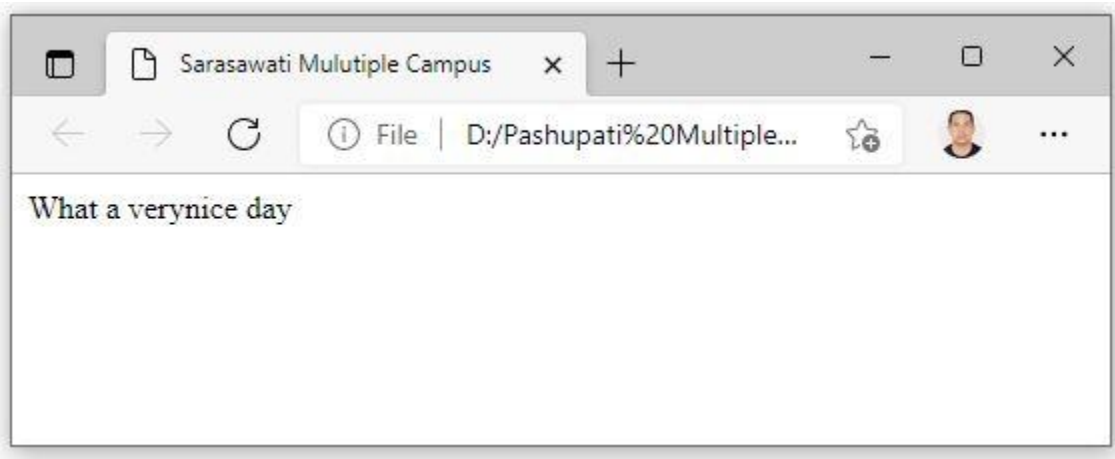
txt2="nice day"

txt3=txt1+txt2

document.writeln(txt3)

```
</script>
</body>
```

Out Put:



Conditional Operator

JavaScript also contains a conditionals operator that assigns a value to a variable to based on some condition.

Syntax:

Variable_name=(condition)? Value-1:Value-2

Example:

```
<body>
```

```
<script type="text/javascript">
```

```
age=20
```

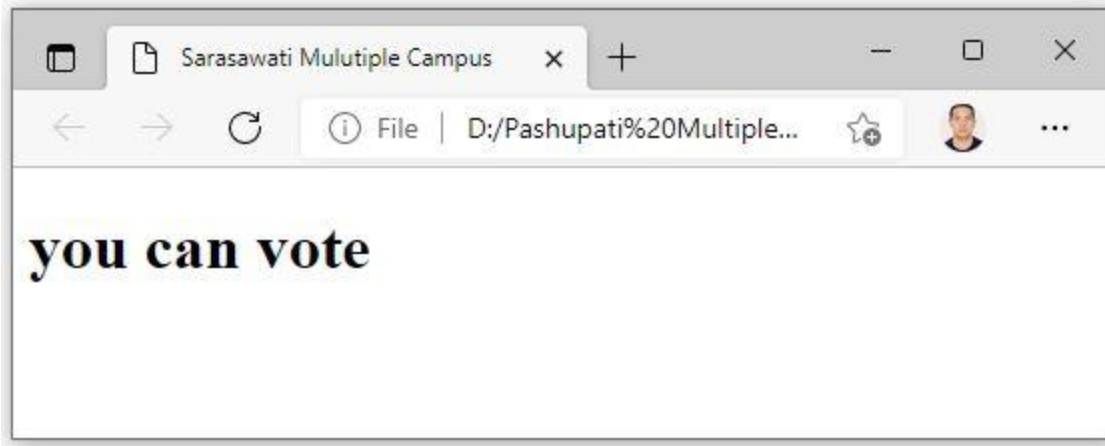
```
check_age=(age>=18)? "you can vote": "You cannot vote"
```

```
document.writeln("<h1>" + check_age + "</h1>")
```

```
</script>
```

```
</body>
```

Out Put:



CONDITIONAL STATEMENT

- ☞ Conditional statement is used to regulate the flow of script.
- ☞ In JavaScript we have the following conditional statements.

1. if statement

- ☞ When the given condition is true it prints the statement; otherwise it skips the statement.

Syntax:

```
if(condition)
{
    //code to be executed if condition is true.
}
```

Example: Write a "Good Morning" greeting if the time is less than 12.

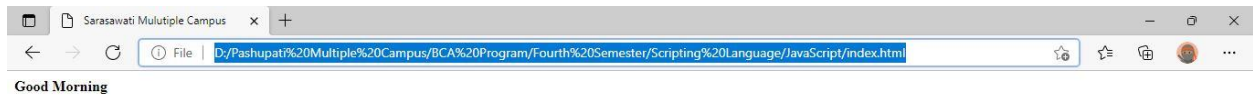
```
<HTML>
<HEAD>
<TITLE></TITLE>

</HEAD>
<BODY>
<script type="text/javascript">
var d= new Date();
var time=d.getHours();
if(time<12)
{
    document.write("<b>Good Morning</b>");
}
</script>

</BODY>
```

</HTML>

Out Put:



2. if-else statement

☞ When the given condition is true if part will be execute otherwise else part will be execute.

Syntax:

```
if(condition)
```

```
{
```

```
    //statement-1
```

```
}
```

```
else
```

```
{
```

```
    //statement-2
```

```
}
```

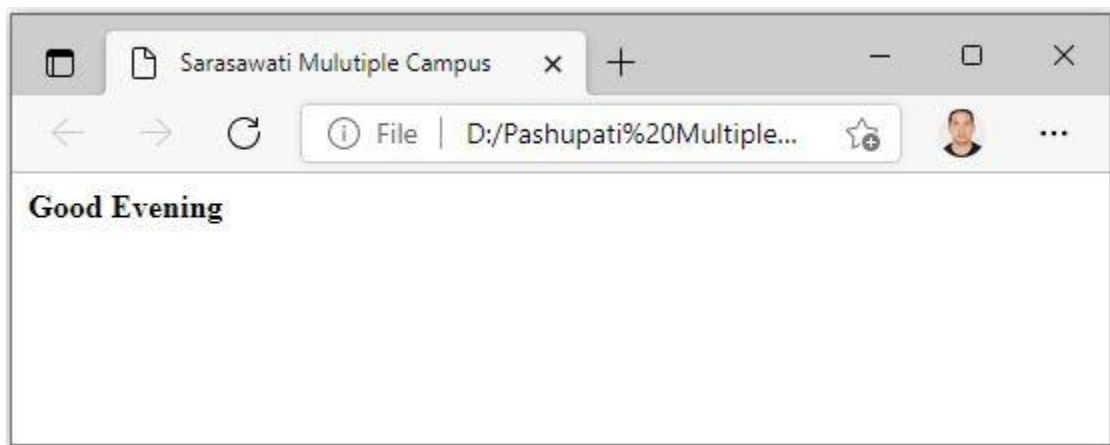
Example:

```
<HTML><head>
```

```
<title>Sarasawati Mulutiple Campus</title>
</head>
<body>
<script type="text/javascript">
var d =new Date();
var time=d.getHours();
if(time<12)
{
    document.write("<b>Good Morning</b>");
}
else
{
    document.writeln("<b>Good Evening</b>");
}

</script>
</body>
</html>
```

Out Put:



3. if...else if... else statement

- ☞ Compound if-else statement print only one statement at a time when more than one condition is given.

Syntax:

```
if(condition-1)
{
    //code to be executed if condition-1 is true.
}
else if(condition-2)
{
    //code to be executed if condition-2 is true.
}
else if(condition-3)
{
    //code to be executed if condition-3 is true.
}
else
{
    //code to be executed if condition-1, condition-2, condition-3 are not true.
}
```

Example

```
<HTML>
<head>
<title>Sarasawati Mulutiple Campus</title>
</head>
<body>
<script type="text/javascript">
var d =new Date();
var time=d.getHours();
if(time<10)
{
    document.write("<b>Good Morning</b>");
}
else if(time>10&& time<16)
{
```

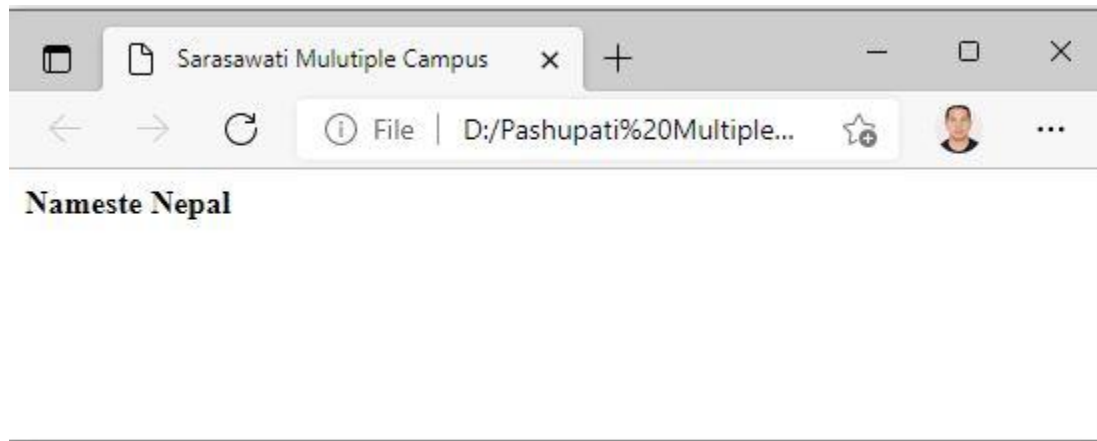
```

        document.writeln("<b>Good Day</b>");
    }
    else
    {
        document.writeln("<b>Nameste Nepal</b>");
    }

</script>
</body>
</html>

```

Out Put:



4. switch statement

- ☞ You should use the switch statement if you want to select one of many block of code to be executed. Conditional statements in JavaScript are used to perform different actions based on different conditions.

Syntax:

```

switch(exp)
{
    case 1:
        execute code bolock-1
        break;
    case 2:
        execute code block-2
        break;
    default:

```

code to be executed if exp is different from case-1 and case-2ss

}

Example:

<body>

<script type="text/javascript">

var d =new Date();

day=d.getDay();

switch (day)

{

case 0:

document.writeln("Today is SUNDAY");

break;

case 1:

document.writeln("Today is MONDAY");

break;

case 2:

document.writeln("Today is TUESDAY");

break;

case 3:

document.writeln("Today is WEDNESDAY");

break;

case 4:

document.writeln("Today is THURSDAY");

break;

case 5:

document.writeln("Today is FRIDAY");

break;

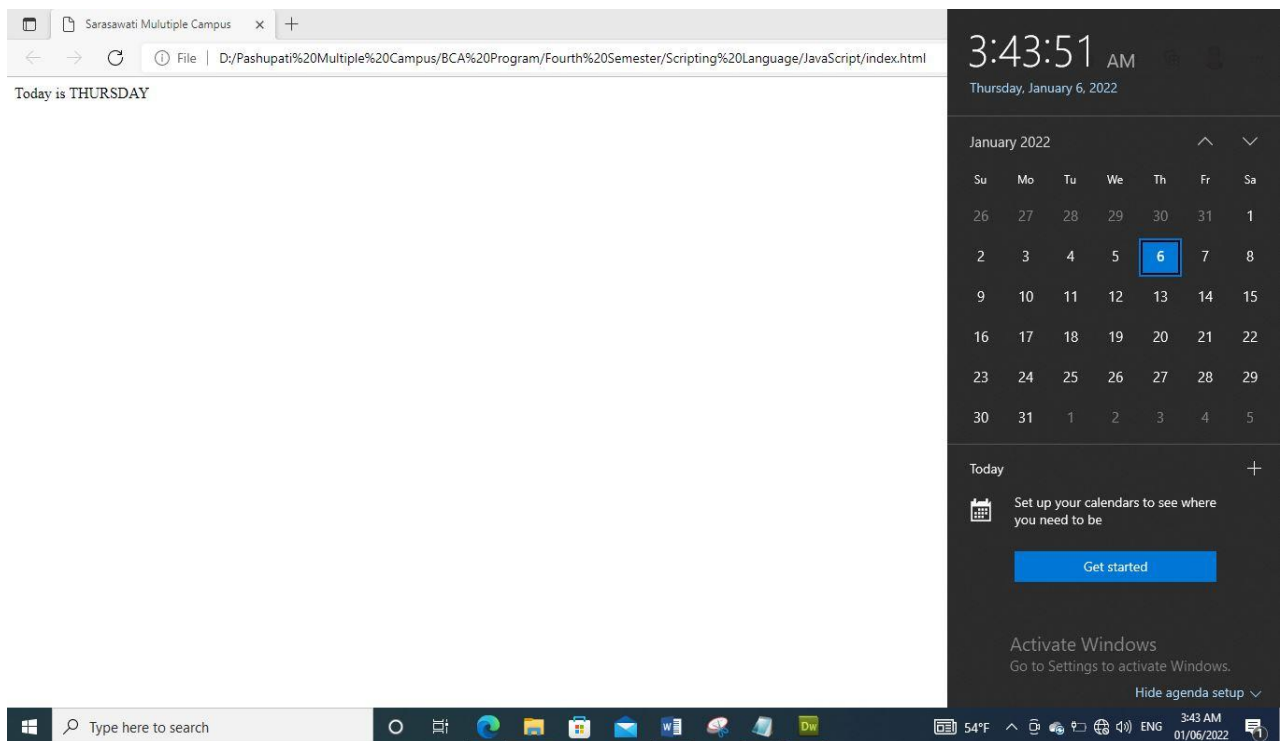
case 6:

```
document.writeln("Today is SATURDAY");  
  
break;  
  
default:  
  
document.writeln("Case does not match");  
  
}
```

</script>

</body>

Out Put:



JavaScript Loop

Loop:

- ☞ When one statement requires number of time then it is said to be a loop.

☞ The different types of loop used in JavaScript are:

- I. while loop
- II. do-while loop
- III. for loop

I. while loop

☞ while loop print the statement after checking its condition.

Syntax:

```
while(var<=endvalue)
{
    //code to be executed
}
```

Example:

```
<title>Sarasawati Mulutiple Campus</title>
```

```
</head>
```

```
<body>
```

```
<script type="text/javascript">
```

```
var i=1
```

```
document.writeln("The number from 1 to 10 is:<br>")
```

```
while(i<=10)
```

```
{
```

```
    document.writeln(""+i)
```

```
    document.writeln("<br>")
```

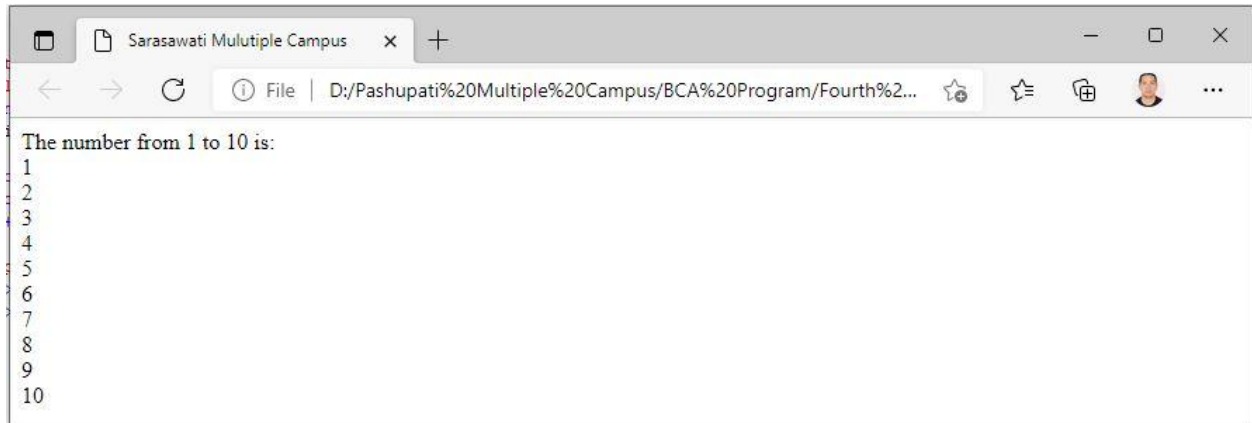
```
    i++;
```

```
}
```

```
</script>
```

```
</body>
```

Out Put:



Exampe-2:

```
<title>Sarasawati Mulutiple Campus</title>
```

```
</head>
```

```
<body>
```

```
<script type="text/javascript">
```

```
var i=1
```

```
var sum=0
```

```
document.writeln("The Sum from 1 to 10 is:<br>")
```

```
while(i<=10)
```

```
{
```

```
    sum=sum+i
```

```
    i++;
```

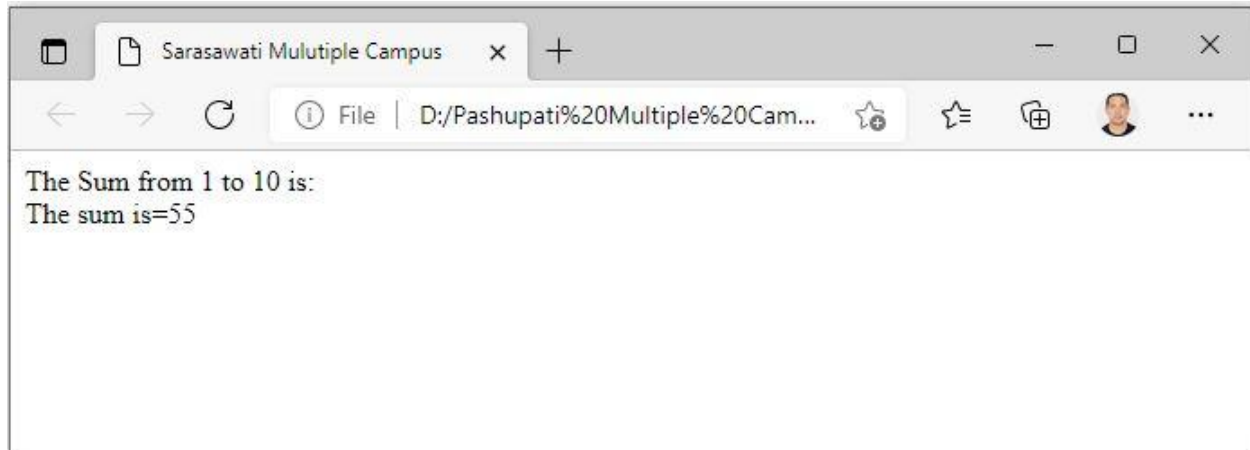
```
}
```

```
document.writeln("The sum is="+ sum)
```

```
</script>
```

```
</body>
```

Out Put:



II. do-while loop

☞ do-while loop print the statement at least one before checking its condition.

Syntax:

```
do
{
    //code to be executed
}while(var<=endvalue);
```

Example:

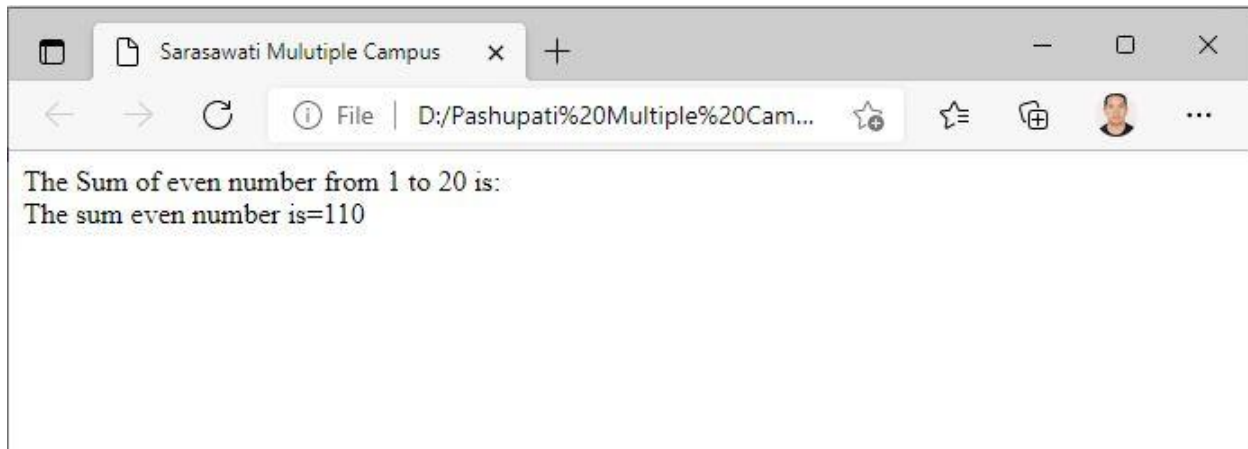
```
<body>
<script type="text/javascript">
var i=2
var sum=0
document.writeln("The Sum of even number from 1 to 20 is:<br>")
do
{
    sum=sum+i
    i=i+2
}while(i<=20);
```

```
document.writeln("The sum even number is="+ sum)
```

```
</script>
```

```
</body>
```

Out Put:



III. for Loop

☞ The for loop is used when you know in advance how many times the script should run.

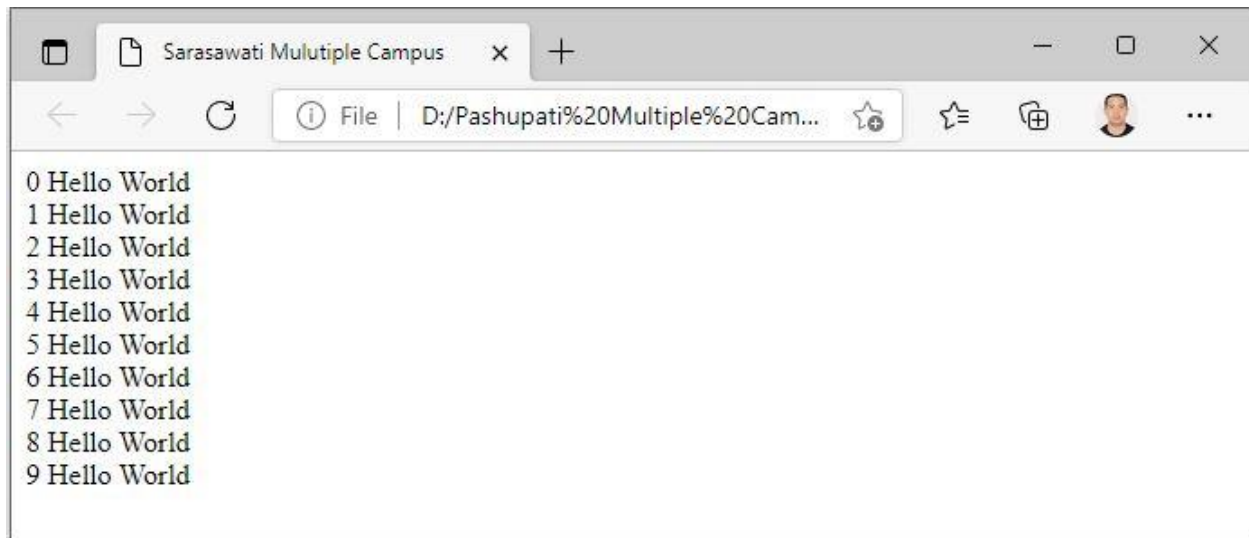
Syntax:

```
for(var=startvalue; var<=endvalue; var=var+increment)
{
    //code to be executed
}
```

Example:

```
<body>
<script type="text/javascript">
var i
for(i=0;i<10;i++)
{
    document.writeln(i)
    document.writeln("Hello World<br>")
}
</script>
</body>
```

Out Put:



Example-2:

```
<body>
```

```
<script type="text/javascript">
```

```
var n=2
```

```
for(i=1;i<=10;i++)
```

```
{
```

```
    var r=n*i
```

```
    document.write(r)
```

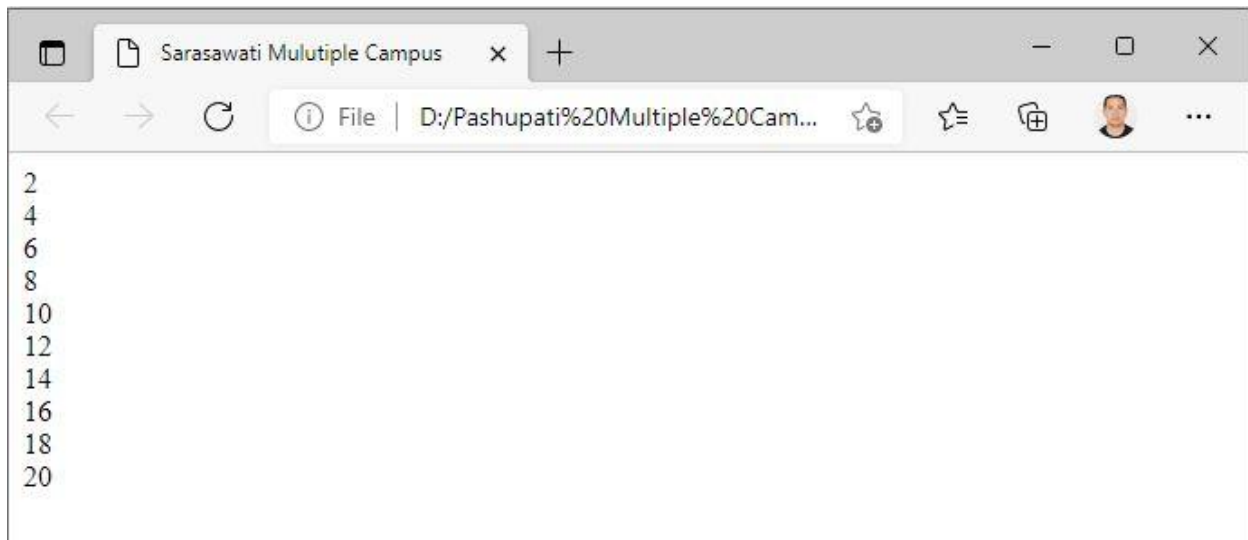
```
    document.writeln("<br>")
```

```
}
```

```
</script>
```

```
</body>
```

Out Put:



Factorial of given number n

```
<body>
```

```
<script type="text/javascript">
```

```
var n=5
```

```
var fact=1
```

```
for(i=n;i>=1;i--)
```

```
{
```

```
    fact=fact*i
```

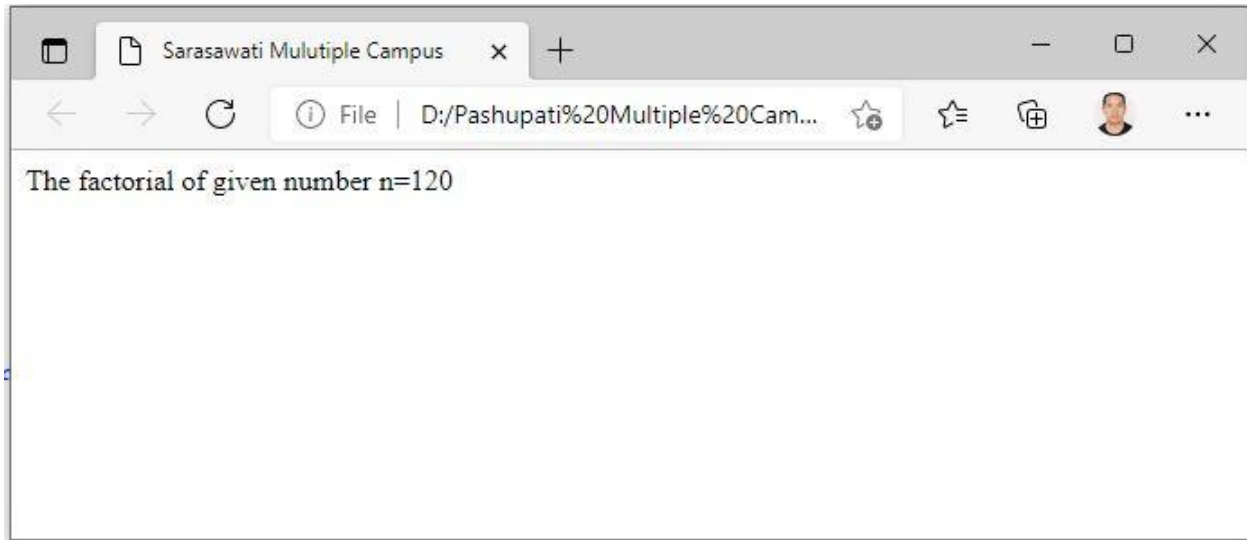
```
}
```

```
document.write("The factorial of given number n="+fact)
```

```
</script>
```

```
</body>
```

Out Put:



Object based Programming with Java Script and Event handling

What is an object?

Object is an entity which have state and behavior (properties and methods).

Properties and methods are bind together in a single unit is called object. For example, car, pen, student, chair, keyboard, monitor etc.

Creating Object in JavaScript

There are three ways to create object in JavaScript:

- ❖ By object literal
- ❖ By creating an instance of Object directly (using **new** keyword)
- ❖ By using object constructor (using a **new** keyword)

- ❖ By object literal

Syntax:

```
Object_name={ property1:value1, property1:value2,property3:value3.....  
Property-n : value-n)
```

Example: Example program to demonstrate object literal

```
<body>
```

```
<script type="text/javascript">
```

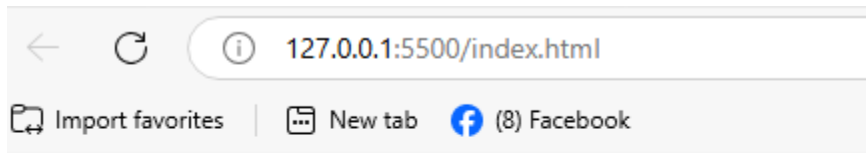
```
student={id:"01",name:"Ramesh",address:"Kathmandu",class1:"twelve"}
```

```
document.writeln(student.id+"&nbsp;&nbsp;&nbsp;"+student.name+"&nbsp;&nbsp;&nbsp;"+student.ad  
dress+"&nbsp;&nbsp;&nbsp;"+student.class1);
```

```
</script>
```

```
</body>
```

Output:



❖ By creating an instance of Object directly (using new keyword)

Syntax:

```
Var objectname=new Object();
```

Example:

```
<body>
```

```
<script type="text/javascript">
```

```
var std=new Object();
```

```
std.id="01";
```

```
std.name="Mahesh Chaudhary";
```

```
std.address="Butwal";
```

```
std.grade="Twelve";
```

```
document.write(std.id+"&nbsp;&nbsp;&nbsp;"+std.name+"&nbsp;&nbsp;&nbsp;"+std.address+"&nbsp;&nbsp;&nbsp;"+std.grade)
```

```
</script>
```

```
</body>
```

Output:



❖ By using object constructor (using a new keyword)

Here we need to create function with arguments. Each argument value can be assigned in the current object by using **this** keyword.

Note: **this** keyword refers to the current object.

Example:

```
<body>

<script type="text/javascript">

function student(id,name,address,class1){
    this.id=id;
    this.name=name;
    this.address=address;
    this.class1=class1;
}

std=new student(1,"Rajan","Biratnagar","Twelve");

document.writeln(std.id+"&nbsp;" +std.name+"&nbsp;" +std.address+"&nbsp;" +std.class1)

</script>

</body>
```

Output:



Predefined Object in JavaScript

- Some important predefined Object in JavaScript are:
 1. Date Object
 2. Number Object
 3. String Object
 4. Array Object
 5. Math Object

Date Object:

The Date object is a datatype built into the JavaScript.

Date object is created with the help of **new Date()**.

Some of the date methods are:

Method	Description
Date()	It is used to return current date and time.
getDate()	Returns the day of the month for the specified date according to local time.
getDay()	Returns the day of the week for the specified date according to local time.
getFullYear()	Returns the year of the specified date.
getHour()	Returns the hour on the specified date.
getMinutes()	Returns the minutes on the specified date.
getMonth()	Returns the month on the specified date.
setDate()	Sets the day of the month for a specified date according to local time.

Example: Write a "Good Morning" greeting if the time is less than 12.

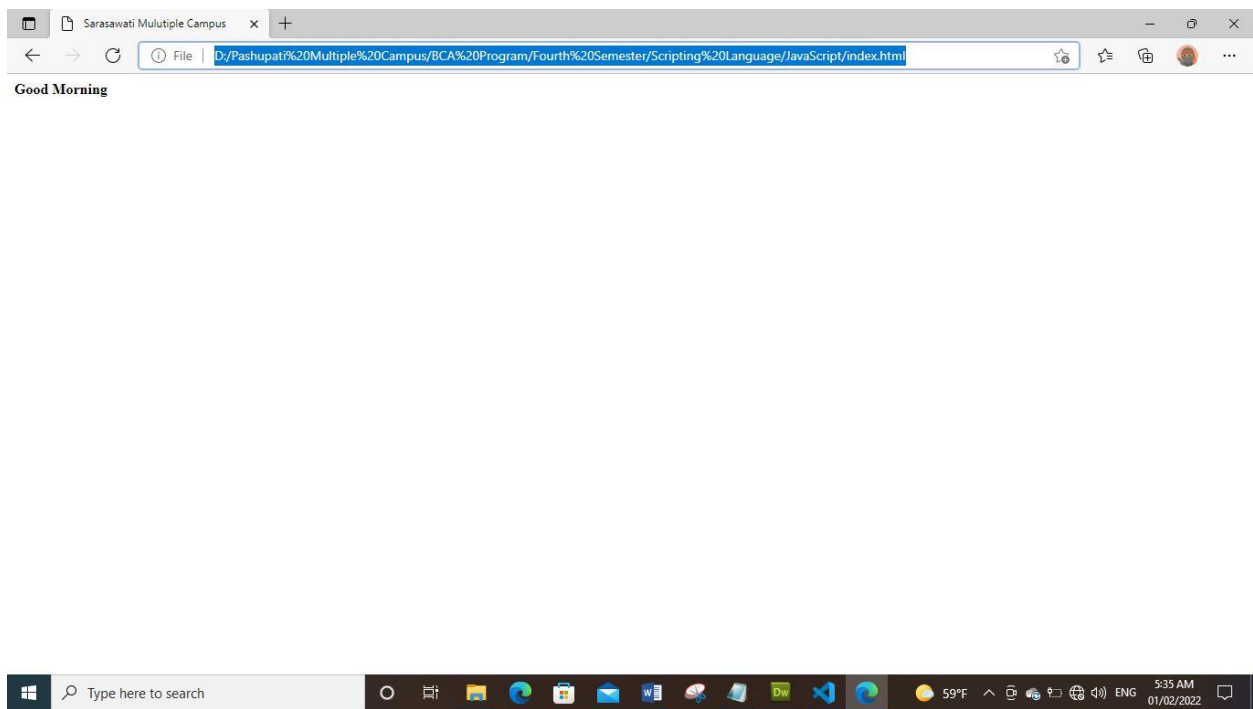
```
<HTML>
<HEAD>
<TITLE></TITLE>

</HEAD>
```

```
<BODY>
<script type="text/javascript">
var d= new Date();
var time=d.getHours();
if(time<12)
{
    document.write("<b>Good Morning</b>");
}
</script>

</BODY>
</HTML>
```

Out Put:



Event Handling:

Some of the events are:

1. onclick event
2. onsubmit
3. onmouseover and onmouseout
4. onfocus and onblur
5. onselect
6. onchange
7. ondblclick

OnClick:

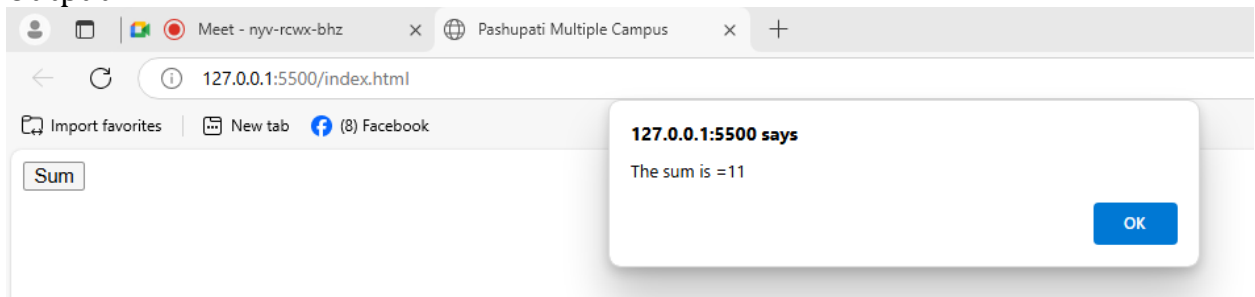
This event occurs when user click the left button of mouse.

Example:

```
<body>
  <script type="text/javascript">
    function sum()
    {
      var a=5;
      var b=6;
      var c=a+b;
      alert("The sum is =" +c)
    }

  </script>
  <form>
    <input type="button" onclick="sum()" value="Sum">
  </form>
</body>
```

Output:



onmouseover and onmouseout

```
<body>
  <script type="text/javascript">
    function sum()
    {
      var a=5;
      var b=6;
      var c=a+b;
      alert("The sum is =" +c)
    }

  </script>
```

```
<form>
  <input type="button" onmouseover="sum()" value="Sum">
</form>
</body>
```

onfocus and onblur

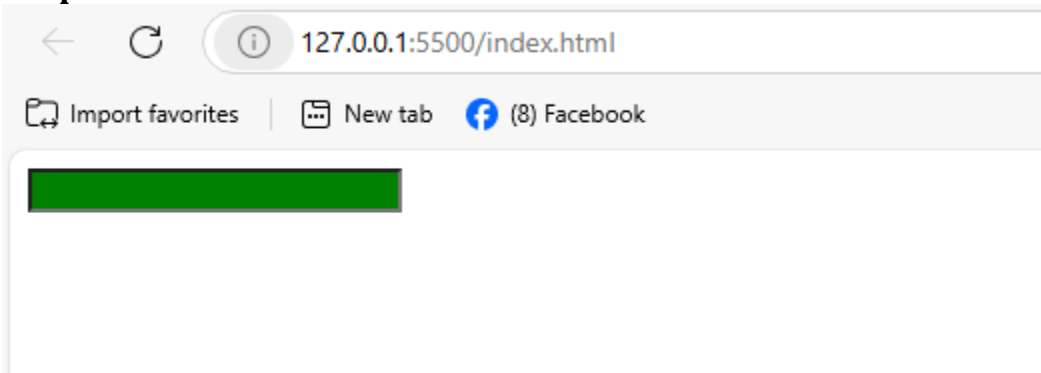
```
<body>
  <form id="form1">
    <input type="text" id="myinput">

  </form>
  <script type="text/javascript">
    var a=document.getElementById("form1");
    a.addEventListener("focus",myFocus,true);
    a.addEventListener("blur",myBlur,true);
    function myFocus(){
      document.getElementById("myinput").style.backgroundColor="red";
    }
    function myBlur(){
      document.getElementById("myinput").style.backgroundColor="";
    }

  </script>

</body>
```

Output:



Form Validation

PHP (Personal Home Page) and Mysql

- ✎ PHP is a server side programming language most of the web application uses PHP like Facebook, tutor , linkedin etc.

Advantages of PHP

- ❖ It is open source programming language.
- ❖ Inexpensive hosting and setup.
- ❖ Cross-Platform (windows, Linux, IOS)
- ❖ Client application does not need PHP installation.
- ❖ Lots of free libraries and packages available for use.

Canonical PHP tag

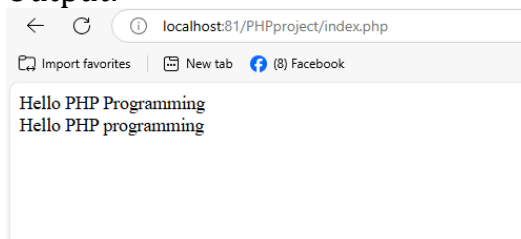
- ✎ PHP start with '<?php' and end with '?>' .
- ✎ Every PHP command end with semicolon (;).

Example: Write PHP program to display statement "Hello PHP Programming".

Code:

```
<?php
echo("Hello PHP Programming<br>");
echo "Hello PHP programming";
?>
```

Output:



How to write comment in PHP

- ❖ Single line comment: The single line comment start with # or // sign.
- ❖ Multiline comment: It start with /* and end with */
Eg/* This is multiline comment*/

PHP Data Type

- ✎ PHP data types are used to hold different types of data or values.

✎ PHP supports 8 primitive data types that can be categorized further in 3 types.

- ❖ Scalar Types (It is also known as predefined)
- ❖ Compound Types (It is also known as user defined type)
- ❖ Special Types

PHP Scalar Data Types

- ✎ It holds only single value.
- ✎ Basically there are four scalar data type in PHP"
 - I. Boolean
 - II. Integer
 - III. Float
 - IV. String

PHP Compound Data Types

- ✎ It can hold multiple values.
- ✎ There are two compound data types in PHP:
 - ❖ Array
 - ❖ Object

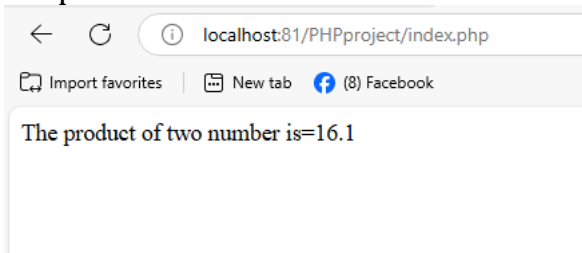
PHP special Data Types

- ✎ There are two special data types in PHP
 - ❖ resource
 - ❖ NULL

Example:

```
<?php
$a=3.5;
$b=4.6;
$p=$a*$b;
echo("The product of two number is=".$p);
?>
```

Output:



PHP Operators

✎ PHP language supports following operators:

- ❖ Arithmetic Operator
- ❖ Comparison Operator
- ❖ Logical or Relational Operator
- ❖ Assignment Operator
- ❖ Conditional Operator

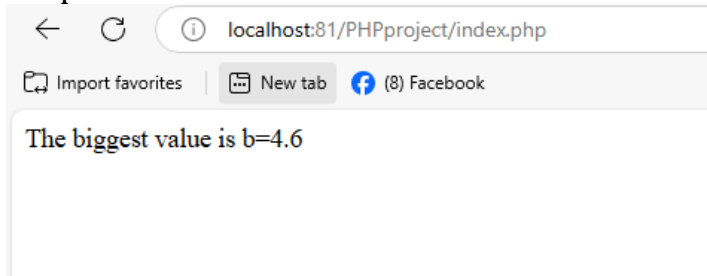
Example:

Write a php program to find biggest among two numbers?

Code:

```
<?php
$a=3.5;
$b=4.6;
if($a>$b)
{
    echo("The biggest value is a=".$a);
}
else{
    echo("The biggest value is b=".$b);
}
?>
```

Output:



Database and Mysql

Database:

Well organized collection of data is known as database.

DBMS (Database Management Software)

DBMS is software which is used to manage data in a database. For example: MySql, Sql-Server, Oracle etc.

How to connect database using PHP

mysqli_connect()

This function is used to connect with MySQL database.

Syntax:

```
$conn=mysqli_connect(server,username,password,databasename);
```

mysqli_close()

This function is used to disconnect with Mysql database.

Syntax:

```
Mysqli_close($conn);
```

Example:

```
<?php
```

```
$server='localhost';
```

```
$user='root';
```

```
$password='';
```

```
$db='test';
```

```
$conn=mysqli_connect($server,$user,$password,$db);
```

```
if($conn)
```

```
{
```

```
    echo("Database Connection successful");
```

```
}
```

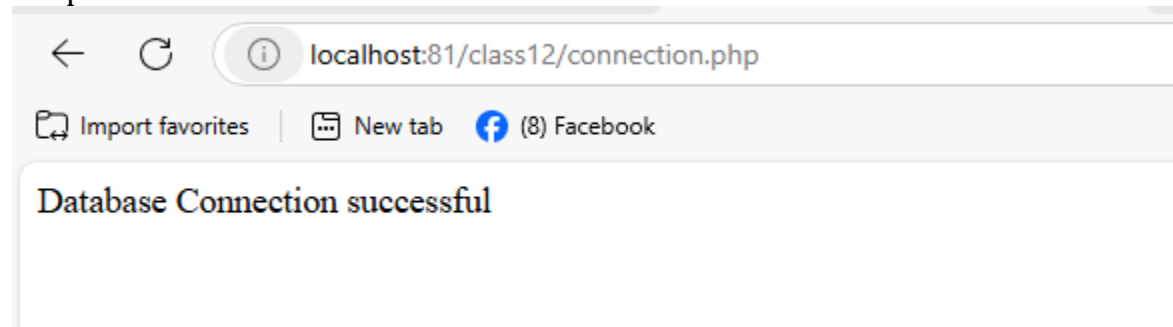
```
else{
```

```
    die("Error in database connection");
```

```
}
```

```
?>
```

Output:



What is SQL Queries?

SQL Queries is a database language which is used to access and manipulate databases.

Some important function of SQL Queries are:

- I. It can execute query against a database.
- II. It can retrieve data from a database.
- III. It can insert, update, delete records from database.
- IV. It can create new database.
- V. It can create new table in a database.
- VI. It can create stored procedures in a database.
- VII. It can create view in a database.
- VIII. It can set permissions on tables, procedures and views data.

SQL Commands

Some important sql commands are:

1. CREATE DATABASE : It is used to create new database.

Syntax:

CREATE DATABASE database_name;

Eg. CREATE DATABASE college;

2. CREATE TABLE : It is used to create new table in a database.

Syntax:

CREATE TABLE table_name(field1,field2,field3.....);

Eg. CREATE TABLE student(

id int(12) primary key auto increment,
name varchar(40),
address varchar(30),
contact varchar(10));

3. INSERT INTO : It is used to insert data into database.

Syntax:

INSERT INTO
table_name(field1,field2,field3.....)Values(value1,value2,value3.....);

Eg. INSERT INTO student(id,name,address,contact)Values(1,'Ram Singh','Kathmandu',98065789');

4. SELECT : It is used to extracts data from a database.

Syntax:

SELECT * FROM table_name ;

Eg. SELECT * FROM student;

5. UPDATE : It use to modify or update data in a database.

Syntax:

```
UPDATE table_name SET some_column = some_value  
WHERE some_column=some_value;
```

Eg.

```
UPDATE student SET address='Biratnagar' WHERE id=1;
```

6. DROP DATABASE : it is used to delete database.

Syntax:

```
DROP DATABASE database_name;  
Eg. DROP DATABASE college;
```

7. DROP TABLE: it is used to delete table from database.

Syntax:

```
DROP TABLE table_name;
```

Eg. DROP TABLE student;

8. ALTER DATABASE: It is used to modify a database

9. ALTER TABLE: It is used to modify table

10.CREATE INDEX : It is used to create index (use for search key)

11.DROP INDEX : It is used to deletes an index.

WHERE Clause:

The WHERE Clause is used to filter records. It is used to extract only those records that satisfy specified condition.

Syntax:

```
SELECT * FROM table_name WHERE condition;
```

Eg.

```
SELECT * FROM students WHERE address='kathmandu';
```

Output:

						id	name	address	contact	email	
☐		Edit		Copy		Delete	1	Ram shrestha	Kathmandu	9841980045	ram@gmail.com
☐		Edit		Copy		Delete	7	Rajendra	kathmandu	987654332	rajendra@gmail.com

View Detail Records:

```
<!DOCTYPE html>
```

```

<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Record Details</title>
</head>
<body>
    <center><font size="24" color="blue">Record Details</font></center>
    <p>&nbsp;&nbsp;&nbsp;</p>
    <hr color="blue">
    <?php
    include 'connection.php';
    if($con->connect_error)
    {
        die("Connection is failed:".$con->connect_error);
    }
    $sql="SELECT * FROM stdrecord";
    $result=$con->query($sql);
    if($result->num_rows>0)
    {
        while($row=$result->fetch_assoc()){
            echo"id:&nbsp;&nbsp;&nbsp;".$row["id"]."Name:&nbsp;&nbsp;&nbsp;".$row["name"]."Ad
dress:&nbsp;&nbsp;&nbsp;".$row["address"]."Contact:&nbsp;&nbsp;&nbsp;".$row["contact"]."Email
:&nbsp;&nbsp;&nbsp;".$row["email"]."<br>";
        }
    }
    else{
        echo"Record not found";
    }
    $con->close();
    ?>
</body>
</html>

```

Output:

Record Details

id: 1Name: ram singhAddress: kathmanduContact: 9876543212Email: ram@gmail.com
id: 2Name: Mahesh ShresthaAddress: PokharaContact: 980675432Email: mahesh@gmail.com
id: 3Name: Binod ChaudharyAddress: BirgunjContact: 346789000Email: binod@gmail.com
id: 4Name: Arjun ShresthaAddress: SindhuliContact: 980076543Email: arjun@gmail.com
id: 5Name: Mahesh ShresthaAddress: kathmanduContact: 980675432Email: mahesth@gmail.com